Soma Yeluri

Software Engineer

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SUMMARY:

Software Engineer with 5+ years of experience designing and developing scalable, secure, event-driven applications using Java, Spring Boot, REST APIs, Kafka, SQL and AWS/Azure. Experienced in building high-volume transaction, payments, healthcare/PBM, claims and customer-service platforms, with a strong focus on API performance, microservices, asynchronous processing and cloud-native architecture. Proven ability to integrate AI/ML capabilities into enterprise workflows, optimize databases and backend services and implement secure solutions aligned with PCI DSS, HIPAA, AML/KYC and other regulatory requirements. Skilled in automated testing, CI/CD, observability and production reliability, with a track record of improving application performance, reducing defects and supporting highly available services.

TECHNICAL SKILLS:

Languages:	Java, SQL, Python, JavaScript, TypeScript, Bash, XML, JSON
Java & Backend:	Java 8/11/17/21, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, JDBC, Spring Cloud, Spring Batch, Maven, Gradle
APIs & Architecture:	REST, SOAP, Microservices, OpenAPI/Swagger, OAuth 2.0, JWT, API Gateway, Design Patterns, System Design, Distributed Systems, DDD
Messaging & Integration:	Apache Kafka, Kafka Streams, RabbitMQ, JMS, Event-Driven Architecture, CDC, SFTP, Batch Processing, Semantic Search, Retrieval & Reranking
AI & GenAI:	OpenAI API, Azure OpenAI, LangChain, LangGraph, RAG, Prompt Engineering, Embeddings, Vector Search, Pinecone, FAISS, AI Agents, Function/Tool Calling, MCP, LLM Integration
Databases:	Oracle, PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Query Optimization
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda, SQS, ECS/EKS, CloudWatch), Azure (Azure DevOps, Azure Monitor, Azure App Service), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD
Testing & Monitoring:	JUnit, Mockito, TestNG, Cucumber, REST, Postman, SonarQube, Splunk, ELK, Prometheus, Grafana, OpenTelemetry
Security:	RBAC, SSO, TLS/SSL, OWASP, Secure API Development, Encryption
Domain:	HIPAA, HITECH, PHI, HL7, FHIR, EDI/X12, Healthcare Claims, Eligibility, Provider, Member, AML, KYC, PCI DSS, SOX, SEC, Payments, Banking, Fraud Detection, Transaction Processing
Tools & Practices:	IntelliJ IDEA, Jira, Confluence, Git, Agile/Scrum, SDLC, Code Review, Performance Tuning

WORK EXPERIENCE:

Cigna Healthcare | USA | Software Engineer | July 2025 – Present

- Designed Java 17/21/Spring Boot microservices for real-time PBM workflows covering member eligibility, pharmacy benefits, prescription coverage, claims, prior authorization and service cases, supporting 2M+ transactions/month.
- Reduced average API response time by ~24% through SQL optimization, indexing, caching, asynchronous processing and JVM tuning across high-volume customer-service services.
- Built secure REST APIs with Spring Security, OAuth 2.0/JWT and OpenAPI, integrating member, pharmacy, claims and prescription systems while supporting 80K+ requests/hour across contact-center channels.
- Integrated LLM-based capabilities into Java/Spring Boot services for claims, eligibility and pharmacy workflows, enabling AI-assisted case summarization and information retrieval.
- Engineered Kafka, SQS/SNS and EventBridge workflows for near-real-time claim, prescription, notification and case updates, processing 1M+ events/day and feeding AI-assisted case-processing workflows with retry and dead-letter recovery.
- Architected AWS services using ECS/EKS, Lambda, API Gateway and Terraform to scale customer-service workloads during demand spikes while maintaining 99.9% API availability.
- Translated PBM business rules for eligibility, formulary, rejected claims, reversals, prior authorization and pharmacy networks into reusable Java components using SQL and JPA/Hibernate.
- Secured PHI/ePHI workflows with encryption, RBAC, least-privilege access, audit logging and data masking, aligning application controls with HIPAA requirements and NCPDP pharmacy transaction workflows.
- Built AI-assisted document processing workflows using Python, REST APIs and AWS, extracting and summarizing claims and service data with PHI masking and RBAC controls.

- Strengthened release quality through JUnit/Mockito testing, Jenkins CI/CD, Docker, CloudWatch monitoring and automated regression validation, reducing production defects by ~18%.

American Express | USA | Software Engineer | Jan 2024 – June 2025

- Owned Java 17/Spring Boot microservices for real-time card authorization, validation, enrichment, reversals and refunds, processing 100K+ events/hour and reducing end-to-end API latency by ~25% through backend optimization.
- Designed Kafka event-driven workflows with partitioning, consumer groups, retries and idempotency, reducing duplicate transaction processing by ~30%.
- Developed fraud-risk services using transaction velocity, merchant, location, device and behavioral signals with Python/scikit-learn, reducing manual fraud-review workload by ~20%.
- Built secure REST APIs connecting transaction, fraud, card/account and notification services using Spring Security, OAuth 2.0, JWT and API Gateway.
- Deployed Dockerized microservices on AWS EKS/Kubernetes with RDS, Redis, CloudWatch, IAM, KMS and Secrets Manager, supporting 99.9% service availability.
- Optimized Oracle/PostgreSQL queries, JPA/Hibernate, indexing and Redis caching, achieving ~22% faster database response times and contributing to the overall API latency improvement.
- Implemented PCI DSS-aligned controls including card-data masking/tokenization, encryption, IAM least privilege, audit logging and secure API practices.
- Developed JUnit/Mockito integration tests and Jenkins CI/CD quality gates, reducing recurring application incidents by ~20% through earlier defect detection and automated validation.

Infosys | India | Software Engineer | June 2020 - Aug 2022

- Developed Java 8/Spring Boot microservices for account servicing, fund transfers, payments and transaction processing, supporting 100K+ daily transaction events.
- Built REST APIs for transaction validation, account/balance checks, authorization, business-rule processing and transaction-status updates, integrating machine-learning-based anomaly scoring for suspicious transaction identification.
- Integrated services with core-banking and payment systems through REST/SOAP APIs, implementing transaction rollback, exception handling and retry processing for failed transactions.
- Integrated Python/scikit-learn anomaly detection with Java/Spring Boot transaction services to flag unusual transaction patterns for review.
- Designed Oracle persistence using SQL, JPA & Hibernate; optimized queries and indexes to improve transaction API response time by ~20%.
- Implemented Apache Kafka producers and consumers for transaction events and asynchronous processing, supporting real-time transaction data pipelines for machine-learning-based fraud and anomaly detection.
- Secured APIs using Spring Security, OAuth 2.0, JWT and RBAC; masked sensitive account/card data, maintained transaction audit trails and supported AML/KYC review and RBI-required banking controls.
- Built pandas/scikit-learn workflows to analyze transaction and account activity, generating risk scores for near-real-time transaction monitoring.
- Deployed and monitored Java services on Microsoft Azure App Service, used Azure Monitor for application health and error tracking and configured Azure DevOps Pipelines for automated build and release, maintaining ~99.9% service availability.
- Automated JUnit/Mockito unit and integration testing within Maven/CI pipelines, increasing regression coverage and reducing recurring production defects by ~25%.

EDUCATION:

Master of Science in Data Science - New Jersey Institute of Technology**, Newark, NJ**
Bachelor of Technology in Computer Science and Engineering - JAIN (Deemed-to-be University)**, Bengaluru, India**